

B

C

Bachelorarbeit

Compileranbindung für die
Programmier-Lern-Plattform
CodeTask

S

Bachelorarbeit

Compileranbindung für die Programmier-Lern-Plattform CodeTask

von

Cilas Handloser

zur Erlangung des akademischen Grades

Bachelor of Science

in Angewandter Informatik

an der Hochschule Konstanz Technik, Wirtschaft und Gestaltung

Matrikelnummer: 307610

Abgabedatum: 30. Juni 2026

Erstbetreuer: **Prof. Dr. Marko Boger**

Zweitbetreuer: **Prof. Dr. Pascal Laube**

Eine elektronische Version dieser Bachelorarbeit ist verfügbar unter <https://github.com/Ciilias/htwg-latex-BA-Cilas>.

Kurzfassung

Abstract. . .

Inhaltsverzeichnis

1	Introduction	1
1.1	Document Structure	1
1.2	Bibliography	2
1.3	Cover and Title Page	3
1.4	Chapters	3
1.5	\section{...}	4
1.5.1	\subsection{...}	4
1.6	Fonts and Colors.	4

1

Introduction

This document is intended to be both an example of the HTWG Konstanz $\text{\LaTeX}$ template for reports and theses, as well as a short introduction to its use. It is not intended to be a general introduction to $\text{\LaTeX}$ itself,¹ and we will assume the reader to be familiar with the basics of creating and compiling documents.

1.1. Document Structure

Since a report, and especially a thesis, might be a substantial document, it is convenient to break it up into smaller pieces. In this template we therefore give every chapter its own file. The chapters (and appendices) are gathered together in `report.tex`, which is the master file describing the overall structure of the document. `report.tex` starts with the line

```
\documentclass{htwg-report}
```

which loads the HTWG Konstanz report template. The template is based on the $\text{\LaTeX}$ book document class and stored in `tudelft-report.cls`. The document class accepts several comma-separated options. The default language is English, but this can be changed to Dutch (e.g., for bachelor theses) by specifying the `dutch` option:

```
\documentclass[german]{htwg-report}
```

¹We recommend <http://en.wikibooks.org/wiki/LaTeX> as a reference and a starting point for new users.

Furthermore, hyperlinks are shown in blue, which is convenient when reader the report on a computer, but can be expensive when printing. They can be turned black with the `print` option. This will also turn the headers black instead of cyan.

If the document becomes large, it is easy to miss warnings about the layout in the $\text{\LaTeX}$ output. In order to locate problem areas, add the `draft` option to the `\documentclass` line. This will display a vertical bar in the margins next to the paragraphs that require attention. Finally, the `nativefonts` option can be used to override the automatic font selection (see below).

This template has the option to automatically generate a cover page with the `\makecover` command. See the next section for a detailed description.

The contents of the report are included between the `\begin{document}` and `\end` commands, and split into three parts by

1. `\frontmatter`, which uses Roman numerals for the page numbers and is used for the title page and the table of contents;
2. `\mainmatter`, which uses Arabic numerals for the page numbers and is the style for the chapters;
3. `\appendix`, which uses letters for the chapter numbers, starting with 'A'.

The title page is defined in a separate file, *e.g.*, `title.tex`, and included verbatim with `\input{title}`.² Additionally, it is possible to include a preface, containing, for example, the acknowledgements. An example can be found in `preface.tex`. The table of contents is generated automatically with the `\tableofcontents` command. Chapters are included after `\mainmatter` and appendices after `\appendix`. For example, `\input{chapter-1}` includes `chapter-1.tex`, which contains this introduction.

1.2. Bibliography

The bibliography, finally, is generated automatically with

```
\printbibliography[heading=bibintoc]
```

from `bib/report.bib`. The bibliography style is specified in `htwg-report.cls`. As an example, we cite the paper by Nobel Prize winner Andre Geim and his pet hamster [\[GT01\]](#). If you need to use a different style, change

```
%% BIB
\RequirePackage[
```

²Note that it is not necessary to specify the file extension.

```

        backend=biber ,
        style=alphabetic ,
        sorting=ynt
]{}bibtex}
\DefineBibliographyStrings{english}{%
    bibliography = {References},
}

```

As compiler, use `biber` to compile and generate the bibliography.

1.3. Cover and Title Page

This template will automatically generate a cover page if you issue the `\makecover` command. However, before generating the cover, you need to provide the information to put on it. This can be done with the following commands:

```

%% 'reporttype' add background elements to the cover / front page
%% possible values are:
%% bachelor      --> B S C
%% master        --> M S C
%% other         --> none
\reporttype{bachelor}

\reporttypetext{Bachelor Thesis}

```

1.4. Chapters

Each chapter has its own file. For example, the $\text{\LaTeX}$ source of this chapter can be found in `chapter-1.tex`. A chapter starts with the command

```
\chapter{Chapter title}
```

This starts a new page, prints the chapter number and title and adds a link in the table of contents. If the title is very long, it may be desirable to use a shorter version in the page headers and the table of contents. This can be achieved by specifying the short title in brackets:

```
\chapter[Short title]{Very long title with many words which could
not possibly fit on one line}
```

Unnumbered chapters, such as the preface, can be created with `\chapter*{Chapter title}`. Such a chapter will not show up in the table of contents or in the page header. To create a table of contents entry anyway, add

```
\addcontentsline{toc}{chapter}{Chapter title}
```

after the `\chapter` command. To print the chapter title in the page header, add

```
\setheader{Chapter title}
```

Chapters are subdivided into sections, subsections, subsubsections, and, optionally, paragraphs and subparagraphs. All can have a title, but only sections and subsections are numbered. As with chapters, the numbering can be turned off by using `\section*{...}` instead of `\section{...}`, and similarly for the subsection.

1.5. `\section{...}`

1.5.1. `\subsection{...}`

`\subsubsection{...}`

`\paragraph{...}` Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

1.6. Fonts and Colors

If you want to use the HTWG house style font Swiss 721 it is necessary to put the font as .ttf file under fonts. As fallback font Arial is used. For more informations to the HTWG house style font see <https://www.htwg-konstanz.de/hochschule/einrichtungen/stabsstelle-kommunikation/corporate-design-logo/>.

Literatur

- [GT01] A. K. Geim und H. A. M. S. ter Tisha. “Detection of earth rotation with a diamagnetically levitating gyroscope”. In: *Physica B: Condensed Matter* 294–295 (2001), S. 736–739. DOI: [10.1016/S0921-4526\(00\)00753-5](https://doi.org/10.1016/S0921-4526(00)00753-5).